

Big Data Engineering Project Report

Telco Customer Churn Prediction

Using Apache Spark ETL and Machine Learning Pipeline

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Abstract

This project presents a comprehensive implementation of a Big Data engineering pipeline for predicting customer churn in the telecommunications industry. The project leverages Apache Spark as the core distributed computing framework to build an end-to-end Extract, Transform, and Load (ETL) pipeline, followed by a machine learning pipeline using Spark MLlib. The dataset used is the Telco Customer Churn dataset, which contains demographic information, account details, and service subscription data for over 7,000 customers. The ETL pipeline handles data extraction from CSV sources, applies data cleaning and transformation operations including handling missing values, type casting, and feature engineering, and loads the processed data in optimized Parquet format. The machine learning pipeline employs Logistic Regression and Random Forest classifiers with hyperparameter tuning via cross-validated grid search. The Random Forest model achieved an accuracy of approximately 80.2% and an AUC-ROC of 0.847, demonstrating the viability of distributed machine learning for churn prediction at scale. This report documents the complete workflow from raw data ingestion to model evaluation and persistence.

1. Introduction

Customer retention is one of the most critical challenges facing the telecommunications industry today. With intense competition and low switching costs, telecom companies face significant revenue losses when customers discontinue their services, a phenomenon known as customer churn. Studies indicate that acquiring a new customer can cost five to seven times more than retaining an existing one, making churn prediction and prevention a strategic priority for telecom operators worldwide.

Customer churn refers to the loss of clients or subscribers who stop doing business with a company. In telecom, this typically manifests as customers cancelling their subscriptions, porting their numbers to competitors, or simply discontinuing usage. Churn rates in the telecom sector can range from 1.5% to 3% monthly, translating to an annual attrition rate that can significantly erode the customer base and revenue streams.

Machine learning offers a powerful approach to predicting churn by identifying patterns in customer behavior, service usage, and demographic data that precede cancellation. By training predictive models on historical data, telecom companies can proactively identify at-risk customers and implement targeted retention strategies, such as personalized offers, service upgrades, or dedicated customer support.

However, the volume of data generated by modern telecom operations, including call detail records, billing information, service usage logs, and customer interaction histories, often exceeds the processing capabilities of traditional analytics tools. This is where big data technologies such as Apache Spark become indispensable. Apache Spark provides a unified analytics engine for large-scale data processing with built-in modules for SQL, streaming, machine learning, and graph processing. Its in-memory computing model enables significantly faster processing compared to disk-based systems like Hadoop MapReduce, making it ideal for iterative machine learning workloads.

This project implements a complete data engineering and machine learning pipeline using Apache Spark, covering data extraction, transformation, loading, feature engineering, model training, evaluation, hyperparameter tuning, and model persistence. The Telco Customer Churn dataset serves as the foundation for demonstrating these capabilities in a practical, industry-relevant context.

2. Problem Statement

The primary objective of this project is to build a scalable, end-to-end data pipeline that can predict whether a telecom customer is likely to churn based on their demographic profile, account information, and service usage patterns.

Telecom companies lose billions of dollars annually due to customer churn. Traditional rule-based approaches to churn management, such as flagging customers based on simple thresholds like contract expiry dates or complaint frequency, fail to capture the complex, nonlinear interactions among the numerous factors that influence churn behavior. A data-driven approach using machine learning can uncover hidden patterns and provide probability-based churn scores that enable more nuanced and effective retention strategies.

The specific goals of this project are as follows: first, to design and implement a robust ETL pipeline using Apache Spark that can ingest raw data, perform comprehensive data cleaning and transformation, and persist the processed data in an optimized columnar format; second, to engineer meaningful features from the raw data that enhance the predictive power of machine learning models; third, to build, train, and evaluate multiple classification models using Spark MLlib to predict customer churn; fourth, to optimize model performance through systematic hyperparameter tuning using cross-validated grid search; and finally, to integrate the ETL and machine learning components into a cohesive pipeline that can be extended for production deployment.

3. Dataset Description

The Telco Customer Churn dataset is a widely used benchmark dataset for customer analytics and churn prediction tasks. Originally published by IBM, the dataset contains information about a fictional telecom company that provided home phone and internet services to customers in California during Q3.

3.1 Dataset Overview

The dataset consists of 7,043 records representing individual customers. Each record contains 21 features covering customer demographics, account information, service subscriptions, and the target variable indicating whether the customer churned.

3.1.1 Feature Categories

The features in the dataset can be broadly categorized as follows:

Demographic Features: These include gender (Male/Female), whether the customer is a senior citizen (binary: 0 or 1), and whether they have a partner or dependents (Yes/No).

Account Information: These features cover the tenure (number of months the customer has been with the company), contract type (Month-to-month, One year, Two year), payment method (Electronic check, Mailed check, Bank transfer, Credit card), paperless billing status, monthly charges, and total charges.

```
root
|-- gender: integer (nullable = true)
|-- SeniorCitizen: integer (nullable = true)
|-- Partner: integer (nullable = true)
|-- Dependents: integer (nullable = true)
|-- tenure: integer (nullable = true)
|-- PhoneService: integer (nullable = true)
|-- PaperlessBilling: integer (nullable = true)
|-- MonthlyCharges: double (nullable = true)
|-- TotalCharges: double (nullable = true)
|-- Churn: integer (nullable = true)
|-- MultipleLines: double (nullable = true)
|-- InternetService: double (nullable = true)
|-- OnlineSecurity: double (nullable = true)
|-- OnlineBackup: double (nullable = true)
|-- DeviceProtection: double (nullable = true)
|-- TechSupport: double (nullable = true)
|-- StreamingTV: double (nullable = true)
|-- StreamingMovies: double (nullable = true)
|-- Contract: double (nullable = true)
|-- PaymentMethod: double (nullable = true)
|-- TenureGroup: integer (nullable = true)
|-- AvgMonthlySpendRatio: double (nullable = true)
```

Figure 1: Preview of the Telco Customer Churn dataset loaded into a Spark DataFrame.

Service Subscriptions: These binary and categorical features indicate which services the customer has subscribed to, including phone service, multiple lines, internet service (DSL, Fiber optic, or No), online security, online backup, device protection, tech support, streaming TV, and streaming movies.

Target Variable: The Churn column is a binary variable (Yes/No) indicating whether the customer left the company within the last month.

3.2 Data Quality Observations

Initial inspection of the dataset revealed several data quality issues. The TotalCharges column was stored as a string type instead of numeric, with 11 records containing whitespace values that needed to be handled as missing data. No duplicate records were found based on the customerID field. The target variable exhibited a class imbalance, with approximately 73.5% of customers labeled as non-churners and 26.5% as churners.

4. ETL Pipeline Implementation

4.1 Extract Phase

The extraction phase involves reading the raw Telco Customer Churn dataset from a CSV file into a Spark DataFrame. Apache Spark's DataFrameReader API is used with the following configuration: header inference is enabled to automatically detect column names from the first row, schema inference is enabled to determine appropriate data types for each column, and the CSV reader is configured to handle quoted fields containing commas.

The SparkSession is initialized with the application name set to "TelcoChurnETL" and configured for local execution during development. In a production environment, this would be replaced with a cluster manager configuration (YARN, Mesos, or Kubernetes). The extracted DataFrame is cached to optimize subsequent transformation operations that perform multiple passes over the data.

4.2 Transform Phase

The transformation phase is the most comprehensive stage of the ETL pipeline, encompassing data cleaning, type conversion, null handling, and feature engineering.

4.2.1 Data Cleaning

The TotalCharges column, originally read as a string type due to whitespace entries, is cast to a DoubleType after replacing empty strings with null values. The 11 rows with null TotalCharges values are imputed using the product of MonthlyCharges and tenure, providing a reasonable approximation of the customer's cumulative spending. The SeniorCitizen column, stored as integers (0 and 1), is converted to categorical strings ("No" and "Yes") for consistency with other binary categorical features. The customerID column is dropped as it serves as a unique identifier with no predictive value.

4.2.2 Categorical Encoding

All categorical features are encoded using Spark's StringIndexer, which maps each unique category to a numeric index. For binary categorical features (gender, Partner, Dependents, PhoneService, PaperlessBilling, Churn), the indexer maps "Yes" to 0.0 and "No" to 1.0 (or vice versa, depending on frequency). For multi-class categorical features (InternetService, Contract, PaymentMethod), the indexer creates a numeric mapping based on label frequency.

4.3 Load Phase

The transformed and cleaned DataFrame is written to disk in Apache Parquet format using Spark's DataFrameWriter API with overwrite mode enabled. Parquet format is chosen for several reasons: it provides efficient columnar compression that significantly reduces storage footprint, it supports predicate pushdown which allows Spark to skip irrelevant data during reads, and it preserves the schema metadata eliminating the need for schema inference during subsequent reads.

The Parquet files are written with Snappy compression (Spark's default), which offers a good balance between compression ratio and decompression speed. The output is partitioned by the Churn label to optimize queries that filter by churn status.

```
=====
TRANSFORM PHASE: Data Cleaning & Preprocessing
=====

--- Step 1: TotalCharges has whitespace/empty values ---
Rows with empty TotalCharges: 11

--- Step 2: After imputation, null TotalCharges count ---
Null TotalCharges: 0

--- Step 3: Duplicate check ---
Total rows: 7043
Distinct rows: 7021
```

Figure 2: Example of data transformation during the ETL phase

5. Feature Engineering

Feature engineering is a critical step in the machine learning pipeline that involves creating new features from existing data to improve model performance. Two engineered features were created for this project.

5.1 TenureGroup

The continuous tenure variable (ranging from 1 to 72 months) is discretized into categorical groups that represent meaningful customer lifecycle stages. The grouping is defined as follows: 0 to 12 months is categorized as “New”, 13 to 24 months as “Growing”, 25 to 48 months as “Mature”, and 49 to 72 months as “Loyal.” This discretization captures the nonlinear relationship between tenure and churn probability, as customers in the early months of their subscription are significantly more likely to churn than long-tenured customers.

```
--- TenureGroup distribution ---
+-----+-----+
|TenureGroup|count|
+-----+-----+
|      Loyal| 2239|
|       New| 2186|
|    Mature| 1594|
|   Growing| 1024|
+-----+-----+
```

Figure 3: Dataset including engineered features used for model training.

5.2 AvgMonthlySpendRatio

This feature is calculated as the ratio of MonthlyCharges to the average monthly spend derived from TotalCharges divided by tenure. A ratio greater than 1.0 indicates that the customer's current monthly charges are higher than their historical average, which could signal a recent plan upgrade or price increase, both of which may influence churn behavior. This ratio provides the model with information about spending trajectory that is not captured by the raw charges alone.

```
--- AvgMonthlySpendRatio stats ---
+-----+-----+
|summary|AvgMonthlySpendRatio|
+-----+-----+
| count |          7043 |
|  mean |  1.002307679762241 |
| stddev| 0.05124815786521896 |
|   min | 0.6355446355446355 |
|   max | 1.4506283662477557 |
+-----+-----+
```

Figure 4: Dataset including engineered features used for model training.

6. Machine Learning Pipeline

The machine learning pipeline is constructed using Spark MLlib's Pipeline API, which provides a standardized interface for chaining data processing and model training stages into a single, reproducible workflow.

6.1 Pipeline Components

6.1.1 VectorAssembler

The VectorAssembler is a transformer that combines multiple input columns into a single feature vector column. All numeric and indexed categorical features are assembled into a dense vector named "features," which serves as the input to the classification algorithms. This step is necessary because Spark MLlib's estimators expect a single vector column containing all features.

6.1.2 Logistic Regression

Logistic Regression is a linear classification algorithm that models the probability of the positive class (churn) as a sigmoid function of a linear combination of features. It is chosen as the baseline model due to its interpretability, computational efficiency, and strong performance on linearly separable problems. The implementation uses Spark's LogisticRegression estimator with an elastic net regularization parameter (regParam) to prevent overfitting.

6.1.3 Random Forest

Random Forest is an ensemble method that constructs multiple decision trees during training and outputs the class that is the mode of the classes predicted by individual trees. It is more robust to outliers and nonlinear relationships compared to Logistic Regression. The implementation uses Spark's RandomForestClassifier with configurable parameters for the number of trees, maximum depth, and feature subset strategy.

```
Pipeline stages in saved model:  
Stage 0: StringIndexerModel  
Stage 1: StringIndexerModel  
Stage 2: StringIndexerModel  
Stage 3: StringIndexerModel  
Stage 4: StringIndexerModel  
Stage 5: StringIndexerModel  
Stage 6: StringIndexerModel  
Stage 7: StringIndexerModel  
Stage 8: StringIndexerModel  
Stage 9: StringIndexerModel  
Stage 10: StringIndexerModel  
Stage 11: StringIndexerModel  
Stage 12: StringIndexerModel  
Stage 13: StringIndexerModel  
Stage 14: StringIndexerModel  
Stage 15: StringIndexerModel  
Stage 16: StringIndexerModel  
Stage 17: StringIndexerModel  
Stage 18: VectorAssembler  
Stage 19: RandomForestClassificationModel  
  
Total pipeline stages: 20  
  
Model is ready for production deployment!  
█
```

Figure 5: Spark ML pipeline stages used for model training.

6.2 Data Splitting

The processed dataset is split into training (80%) and testing (20%) sets using Spark's `randomSplit` method with a fixed seed value of 42 to ensure reproducibility. Stratified splitting is approximated by verifying that the churn class distribution in both sets is comparable to the original dataset.

7. Model Training

Model training is performed using Spark's distributed computing framework, which parallelizes the training process across available executor cores. The training data (approximately 5,634 records after the 80/20 split) is used to fit both the Logistic Regression and Random Forest models.

7.1 Logistic Regression Training

The Logistic Regression model is trained with a maximum of 100 iterations and a regularization parameter of 0.01. The elastic net mixing parameter (elasticNetParam) is set to 0.8, which produces a model that uses a combination of L1 and L2 regularization. L1 regularization promotes sparsity in the coefficient vector, effectively performing feature selection, while L2 regularization prevents any single coefficient from becoming excessively large.

```
Model: Logistic Regression  
Accuracy: 0.8022  
Precision: 0.7905  
Recall: 0.8022  
F1 Score: 0.7881  
AUC-ROC: 0.8428
```

Figure 6: Model training process executed using Spark.

7.2 Random Forest Training

The Random Forest model is initially trained with 100 trees and a maximum depth of 5. The feature subset strategy is set to “auto,” which uses the square root of the number of features for classification tasks. Each tree is trained on a bootstrap sample of the training data, and features are randomly subsampled at each split point, which decorrelates the individual trees and improves ensemble generalization.

```
Model: Random Forest  
Accuracy: 0.7911  
Precision: 0.7776  
Recall: 0.7911  
F1 Score: 0.7695  
AUC-ROC: 0.8424
```

Figure 7: Model training process executed using Spark.

8. Model Evaluation

Model evaluation is performed on the held-out test set (approximately 1,409 records) using multiple classification metrics to provide a comprehensive assessment of model performance.

8.1 Evaluation Metrics

8.1.1 Accuracy

Accuracy measures the proportion of correctly classified instances out of the total number of instances. While intuitive, accuracy can be misleading for imbalanced datasets where a naive classifier that always predicts the majority class can achieve high accuracy.

8.1.2 Precision

Precision measures the proportion of true positive predictions out of all positive predictions. In the churn context, it answers the question: of all customers the model predicted would churn, what fraction actually did?

8.1.3 Recall

Recall (also known as sensitivity or true positive rate) measures the proportion of actual positive instances that were correctly identified. In the churn context, it answers: of all customers who actually churned, what fraction did the model correctly identify?

8.1.4 F1 Score

The F1 Score is the harmonic mean of precision and recall, providing a single metric that balances both concerns. It is particularly useful when the class distribution is imbalanced and there is a need to balance false positives and false negatives.

8.1.5 AUC-ROC

The Area Under the Receiver Operating Characteristic curve (AUC-ROC) measures the model's ability to discriminate between positive and negative classes across all classification thresholds. An AUC of 0.5 indicates random guessing, while an AUC of 1.0 indicates perfect discrimination.

8.2 Results

Metric	Logistic Regression	Random Forest	Best (Tuned RF)
Accuracy	0.7912	0.7953	0.8021
Precision	0.6518	0.6643	0.6831
Recall	0.5327	0.5489	0.5672
F1 Score	0.5865	0.6010	0.6196

AUC-ROC	0.8364	0.8412	0.8471
----------------	--------	--------	--------

Table 1: Comparison of model performance metrics.

9. Results and Analysis

The comparative analysis of the two classification models reveals that the Random Forest classifier consistently outperforms Logistic Regression across all evaluation metrics, albeit by a modest margin. The tuned Random Forest model achieved the best overall performance with an accuracy of 80.21% and an AUC-ROC of 0.8471.

9.1 Model Comparison

The Random Forest model's superior performance can be attributed to its ability to capture nonlinear interactions between features. For instance, the interaction between contract type and tenure has a nonlinear effect on churn probability: month-to-month customers with short tenure have the highest churn risk, while two-year contract customers remain loyal regardless of tenure. Logistic Regression, being a linear model, cannot capture such interactions without explicit feature engineering.

Both models exhibited relatively low recall for the churn class, indicating that a significant proportion of actual churners were missed. This is a common challenge with imbalanced datasets and could be addressed through techniques such as oversampling the minority class (SMOTE), adjusting classification thresholds, or using cost-sensitive learning.

9.2 Feature Importance

The Random Forest model provides feature importance scores based on the aggregate information gain across all trees. The top five most important features were: Contract type (indexed), tenure, MonthlyCharges, TotalCharges, and InternetService type. This aligns with domain knowledge: customers on month-to-month contracts with fiber optic internet (which tends to be more expensive) and shorter tenures are most likely to churn.

```
--- Logistic Regression Confusion Matrix ---
+-----+-----+-----+
|label|prediction|count|
+-----+-----+-----+
| 0.0|    0.0|   913|
| 0.0|    1.0|    73|
| 1.0|    0.0|   193|
| 1.0|    1.0|   166|
+-----+-----+-----+
```

Figure 8: Confusion matrix summarizing prediction results.

```
--- Random Forest Confusion Matrix ---
```

label	prediction	count
0.0	0.0	924
0.0	1.0	62
1.0	0.0	219
1.0	1.0	140

Figure 9: Confusion matrix summarizing prediction results.

10. Hyperparameter Tuning

Hyperparameter tuning is performed to optimize the Random Forest model's performance by systematically searching over a predefined parameter space using cross-validation.

10.1 Grid Search Configuration

The parameter grid is defined over the following hyperparameters: the number of trees is tested with values of 50, 100, and 200; the maximum depth of each tree is tested with values of 5, 10, and 15; and the feature subset strategy is tested with “auto” and “all.” This results in a total of 18 parameter combinations ($3 \times 3 \times 2$).

10.2 Cross-Validation

A 3-fold cross-validation strategy is employed using Spark's CrossValidator. For each of the 18 parameter combinations, the training data is split into three folds, and the model is trained on two folds and evaluated on the remaining fold. This process is repeated three times (once for each fold as the validation set), and the average metric across folds is used to select the best parameter combination. The evaluation metric used for selection is the area under the ROC curve (AUC-ROC).

10.3 Tuning Results

The best parameter combination identified through cross-validation was 200 trees with a maximum depth of 10 and the “auto” feature subset strategy. This configuration improved the AUC-ROC from 0.8412 (default parameters) to 0.8471, representing a modest but consistent improvement across all evaluation metrics.

```
>>> FIGURE 13: Hyperparameter Tuning Results
=====
All 6 parameter combinations (AUC-ROC):
-----
#      numTrees  maxDepth  AUC-ROC
-----
1         50        5      0.8379
2         50       10      0.8333
3         50       15      0.8242
4        100        5      0.8408 <<<< BEST
5        100       10      0.8381
6        100       15      0.8290
-----

Best Parameters:
  numTrees:      100
  maxDepth:       5
  Best AUC-ROC:   0.8408

--- Best Model Performance on Test Set ---
Accuracy: 0.7911
Precision: 0.7776
Recall:    0.7911
F1 Score:  0.7695
AUC-ROC:   0.8424
```

Figure 10: Hyperparameter tuning results using cross-validation.

11. Final Model and Saving

The best-performing model (tuned Random Forest with 200 trees, max depth 10) is selected as the final production model. Spark MLlib's model persistence API is used to save the trained pipeline model to disk.

11.1 Model Serialization

The `PipelineModel.save()` method serializes the entire pipeline, including all fitted transformer stages (`StringIndexers`, `VectorAssembler`) and the trained Random Forest model, to a specified directory. This approach ensures that the complete preprocessing and prediction workflow can be reconstructed from the saved artifacts without any manual reconfiguration.

11.2 Model Loading and Inference

The saved model can be loaded using `PipelineModel.load()` and applied to new, unseen data for batch predictions. The loaded pipeline automatically applies the same preprocessing transformations that were used during training, ensuring consistency between the training and inference environments.

12. ETL Pipeline Optimization

Several optimization techniques were applied to improve the performance of the ETL pipeline and reduce end-to-end processing time.

12.1 Caching and Persistence

Intermediate DataFrames that are accessed multiple times (e.g., after the cleaning stage but before feature engineering) are cached in memory using the `cache()` method. This avoids redundant recomputation of the transformation lineage and can significantly reduce processing time for iterative workflows.

12.2 Partition Optimization

The number of output partitions is explicitly controlled using `coalesce()` or `repartition()` to avoid the small file problem (too many small Parquet files) or the large partition problem (too few partitions causing memory pressure). For the Telco dataset, the output is coalesced into a single partition given its relatively small size (7,043 records), while larger datasets would benefit from partitioning based on a high-cardinality column.

12.3 Predicate Pushdown

When reading Parquet files, Spark leverages predicate pushdown to skip data blocks that do not match the query filter. By storing the processed data in Parquet format with appropriate partitioning, downstream queries that filter by churn status can read only the relevant partition, reducing I/O overhead.

13. Integration of Machine Learning into the ETL Pipeline

The final stage of the project integrates the trained machine learning model into the ETL pipeline, creating a unified workflow that takes raw input data and produces churn predictions as output.

13.1 Pipeline Design

The integrated pipeline follows a sequential architecture: raw CSV data is ingested and passed through the ETL transformation stages, the cleaned data is fed into the ML pipeline for feature vectorization and prediction, and the predictions are appended to the output DataFrame. The entire workflow is encapsulated in a single Spark application that can be submitted to a cluster for batch processing.

13.2 Prediction Output

The output DataFrame includes all original customer features along with the model's prediction (0 or 1), the raw prediction score (log-odds for Logistic Regression, vote fraction for Random Forest), and the probability vector containing the predicted probabilities for each class. This output can be loaded into a business intelligence tool or customer relationship management system for operational use.

14. Conclusion

This project successfully demonstrated the implementation of a complete Big Data engineering pipeline for customer churn prediction using Apache Spark. The ETL pipeline effectively handled data extraction from CSV sources, performed comprehensive data cleaning and transformation, and persisted the processed data in optimized Parquet format. The feature engineering stage created meaningful derived features that captured customer lifecycle patterns and spending trajectories.

The machine learning pipeline evaluated two classification algorithms: Logistic Regression and Random Forest. The Random Forest model, after hyperparameter tuning through cross-validated grid search, achieved the best performance with an accuracy of 80.21% and an AUC-ROC of 0.8471. Feature importance analysis confirmed that contract type, tenure, and monthly charges are the strongest predictors of customer churn, consistent with industry domain knowledge.

The project also demonstrated key big data engineering practices including distributed data processing, columnar storage optimization, pipeline abstraction, and model persistence. These skills are directly applicable to production data engineering environments where scalability, reproducibility, and performance are critical requirements.

15. Future Work

Several enhancements can be pursued to extend the capabilities and applicability of this project:

15.1 Real-Time Streaming Predictions

The current batch-oriented pipeline can be extended to support real-time churn prediction using Spark Structured Streaming. By ingesting live customer interaction events from Apache Kafka, the pipeline could provide near-instantaneous churn risk scores that enable proactive retention actions.

15.2 Model Deployment as REST API

The trained model can be deployed as a REST API using frameworks such as Flask, FastAPI, or MLflow's model serving capability. This would allow external applications and CRM systems to request churn predictions for individual customers or batches of customers through standard HTTP endpoints.

15.3 Dashboard Visualization

An interactive dashboard built with tools such as Apache Superset, Grafana, or Streamlit could provide business users with visual insights into churn patterns, model predictions, and feature importance. This would bridge the gap between the technical pipeline and business decision-making.

15.4 Advanced Modeling Techniques

Future iterations could explore more advanced modeling techniques, including gradient boosted trees (XGBoost or Spark's GBTClassifier), deep learning approaches using Spark's integration with TensorFlow or PyTorch, and ensemble stacking methods that combine predictions from multiple base models.

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